```
# question 1

price = 1000;
x = 1000*(20/100);
y = price - x;
price1 = y;
print("The price after 1st discount is ",price1);

z = (10/100)*price1;
price2 = price1 - z;
print("The price after 2nd discount is ",price2);

final_price = price2;
final_discount = ((z+x)/price)*100;
print("The overall discount of the shirt is ",final_discount,"%");
```

```
The price after 1st discount is  800.0
The price after 2nd discount is  720.0
The overall discount of the shirt is  28.000000000000004 %
```

```
# question 2a

item1 = 50.78 ;
item2 = 100.89 ;
x = int(input("Enter the number of item 1 : \n"));
print(x);
y = int(input("Enter the number of item 2 : \n"));
print(y);


price = x*item1 + y*item2;
print("The price before tax is :",price);

z = price*(18/100);
final_price = price + z;
print("The final amount should be paid is :",final_price);
```

```
Enter the number of item 1 :
 2

2

Enter the number of item 2 :
 3

3
The price before tax is : 404.23
The final amount should be paid is : 476.9914
```

```python
# question 2b

m = int(input("Enter how much money Mr. Jones has :"));

item1 = 50.78 ;
item2 = 100.89 ;
x = int(input("Enter the number of item 1 : \n"));
print(x);
y = int(input("Enter the number of item 2 : \n"));
print(y);

price = x*item1 + y*item2;
print("The price before tax is :",price);

z = price*(18/100);
final_price = price + z;
print("The final amount should be paid is :",final_price);

left = m - final_price;
print("money left with Mr. John is :",left);
```

```
Enter how much money Mr. Jones has : 5
Enter the number of item 1 :
 9

9

Enter the number of item 2 :
 9

9
The price before tax is : 1365.03
The final amount should be paid is : 1610.7354
money left with Mr. John is : -1605.7354
```

```python
# 1A

even_no = ['2','4','6','8','10','12','14','16','18','20']
print("The fifth Number is : ",even_no[4])

sub_list = even_no[3:7]
print("The sublist is :",sub_list)
```

```
The fifth Number is :  10
The sublist is : ['8', '10', '12', '14']
```

```python
#1B.

positive_integers = ['1','2','3','4','5']
odd_numbers = ['1','3','5','7','9']

positive_integers.extend(odd_numbers)
print(positive_integers)
```

```
['1', '2', '3', '4', '5', '1', '3', '5', '7', '9']
```

```python
#2A.

my_list = [10, 20, 30, 40, 50, 60, 70, 80, 90, 100]
n = int(input("Enter the position of the element to remove (from the
reverse order): "))

my_list.pop(-n)
print("Updated list:", my_list)
```

```
Enter the position of the element to remove (from the reverse order):
2

Updated list: [10, 20, 30, 40, 50, 60, 70, 80, 100]
```

```python
#2B.

my_list = [10, 20, 30, 40, 50, 60, 70, 80, 90, 100]
element = int(input("Enter the element to search for: "))
get = 0

for item in my_list :
    if item==element :
        get  = 1
if get == 1:
    print(f"The position of {element} in the list is
{my_list.index(element)+1}")
else :
    print(f"{element} is not present in the list")
```

```
Enter the element to search for:  10
```

```
The position of 10 in the list is 1
```

```python
#3A.

my_list = [10, 20, 30, 40, 50, 60, 70, 80, 90, 100]

start_index = int(input("Enter the start index: "))
end_index = int(input("Enter the end index: "))

del my_list[start_index:end_index]
print("Updated list:", my_list)
```

```
Enter the start index:  1
Enter the end index:  6

Updated list: [10, 70, 80, 90, 100]
```

```python
#3B

my_list = [10, 20, 30, 40, 50, 60, 70, 80, 90, 100]

start_index = int(input("Enter the start index: "))
end_index = int(input("Enter the end index: "))

del my_list[start_index:end_index]
print("Updated list:", my_list)
print("Length of the list:", len(my_list))
```

```
Enter the start index:  3
Enter the end index:  5

Updated list: [10, 20, 30, 60, 70, 80, 90, 100]
Length of the list: 8
```

```python
#1

n = int(input("Enter the number of integers: "))

target_tuple = tuple(i + 2 for i in range(n))

print("Generated tuple:", target_tuple)
```

```
Enter the number of integers:  4

Generated tuple: (2, 3, 4, 5)
```

```python
#2


n = int(input("Enter the number of elements in tuple: "))
tup = tuple(int(input(f"Enter element {i+1}: ")) for i in range(n))

min_val = tup[0]
max_val = tup[0]

for i in tup:
    if i < min_val:
        min_val = i
    if i > max_val:
        max_val = i

print(f"Minimum value in the tuple: {min_val}")
print(f"Maximum value in the tuple: {max_val}")
```

```
Enter the number of elements in tuple:  3
Enter element 1:  1
Enter element 2:  2
Enter element 3:  3

Minimum value in the tuple: 1
Maximum value in the tuple: 3
<class 'tuple'>
```

```python
#3.a


m = int(input("Enter the number of elements in T1: "))
T1 = tuple(int(input(f"Enter element {i+1} for T1: ")) for i in
range(m))

n = int(input("Enter the number of elements in T2: "))
T2 = tuple(int(input(f"Enter element {i+1} for T2: ")) for i in
range(n))

T3 = T1 + T2
```

```
list_count = []

for element in T3:
    count = 0
    for x in T3:
        if element == x:
            count += 1
    counts.append(list_count)

print(f"Merged tuple T3: {T3}")
print(f"Occurrences of each element in T3: {counts}")
```

```
Enter the number of elements in T1:  4
Enter element 1 for T1:  12
Enter element 2 for T1:  34
Enter element 3 for T1:  54
Enter element 4 for T1:  1
Enter the number of elements in T2:  2
Enter element 1 for T2:  5
Enter element 2 for T2:  1
```

```
Merged tuple T3: (12, 34, 54, 1, 5, 1)
Occurrences of each element in T3: [2, 2, 1, 1, 2, 2, 1, [], [], [],
[], [], []]
```

```
#3.b

m = int(input("Enter the number of elements in T1: "))
T1 = tuple(int(input(f"Enter element {i+1} for T1: ")) for i in
range(m))

n = int(input("Enter the number of elements in T2: "))
T2 = tuple(int(input(f"Enter element {i+1} for T2: ")) for i in
range(n))

T3 = T1 + T2

T3_unique_sorted = sorted(set(T3))

print(f"Sorted tuple with duplicates removed: {T3_unique_sorted}")
```

```
Enter the number of elements in T1:  3
Enter element 1 for T1:  1
Enter element 2 for T1:  2
Enter element 3 for T1:  3
Enter the number of elements in T2:  3
Enter element 1 for T2:  2
Enter element 2 for T2:  3
Enter element 3 for T2:  4
```

```
Sorted tuple with duplicates removed: [1, 2, 3, 4]
```

```python
#1
def add_new():
    dictionary = eval(input("Enter the dictionary : "))
#like {'name' : 'kishan', 'designation' : 'topper'}

    key = input("Enter the key to add: ")
    value = input("Enter the value for the key: ")

    dictionary[key] = value
    print("Updated dictionary:", dictionary)

add_new()
```

```
Enter the dictionary :   {'name' : 'kishan', 'designation' : 'topper'}
Enter the key to add:  roll no
Enter the value for the key:  csb23010

Updated dictionary: {'name': 'kishan', 'designation': 'topper', 'roll
no': 'csb23010'}
```

```python
#2

def filter_dict(dictionary, threshold):
    filtered_dict = {}
    for key, value in dictionary.items():
        if value>threshold:
            filtered_dict[key] = value;
    return filtered_dict;

my_dict = {
    'x' : 100,
    'y' : 200,
    'z' : 300,
}
result = filter_dict(my_dict, 150)
print("The filtered dictionary is :",result)
```

```
The filtered dictionary is : {'y': 200, 'z': 300}
```

```python
#3

def remove_dict(dictionary):
    if dictionary:
        dictionary.popitem()
        print("The updated dictionary is :", dictionary)
    else:
        print("The dictionary is empty.");

my_dict = {
    'name' : 'Raktim Ranjan Handique',
    'dept' : 'Computer Science and Engineering',
```

```
    'roll no' : 'csb23017'
}
remove_dict(my_dict)
```

```python
#1

def add_new(dictionary, key, value):
    dictionary[key] = value;
    print("The updated dictionary : ",dictionary);

my_dict = {
    'name' : 'Raktim Ranjan Handique',
    'dept' : 'Computer Science and Engineering',
}
add_new(my_dict,'Roll no', 'csb23017')
```

```
The updated dictionary :  {'name': 'Raktim Ranjan Handique', 'dept':
'Computer Science and Engineering', 'Roll no': 'csb23017'}
```

```python
#2

def filter_dict(dictionary, threshold):
    filtered_dict = {}
    for key, value in dictionary.items():
        if value>threshold:
            filtered_dict[key] = value;
    return filtered_dict;

my_dict = {
    'x' : 100,
    'y' : 200,
    'z' : 300,
}
result = filter_dict(my_dict, 150)
print("The filtered dictionary is :",result)
```

```
The filtered dictionary is : {'y': 200, 'z': 300}
```

```python
#3

def remove_dict(dictionary):
    if dictionary:
        dictionary.popitem()
        print("The updated dictionary is :", dictionary)
    else:
        print("The dictionary is empty.");

my_dict = {
    'name' : 'Raktim Ranjan Handique',
    'dept' : 'Computer Science and Engineering',
    'roll no' : 'csb23017'
}
remove_dict(my_dict)
```

```
The updated dictionary is : {'name': 'Raktim Ranjan Handique', 'dept':
'Computer Science and Engineering'}
```

```python
#4

def find_palindromes():

    strings = input("Enter a list of strings separated by spaces:
").split()
    palindromes = []

    for s in strings:
        if s == s[::-1]:
            palindromes.append(s)

    if palindromes:
        return palindromes
    else:
        return "No palindrome"

result = find_palindromes()
print(result)
```

```
Enter a list of strings separated by spaces:  madam hello

['madam']
```

```python
#4

def char_frequency():
    string = input("Enter a string: ")

    frequency_dict = {}
    for char in string:
        frequency_dict[char] = frequency_dict.get(char, 0) + 1

    return frequency_dict

result = char_frequency()
print(result)
```

```
Enter a string:  aaaabbbbccc

{'a': 4, 'b': 4, 'c': 3}
```

```python
#Q1

def find_unique_pair(arr, target_sum):
    pairs = set();
    appeared_numbers = set();

    for num in arr:
        complement = target_sum - num
        if complement in appeared_numbers:
            pairs.add(tuple(sorted((num, complement))))
        appeared_numbers.add(num)
    return list(pairs)

arr = [1, 8, 2, 7, 9, 0, 9, 6]
target_sum = 8
print(find_unique_pair(arr, target_sum))
```

```
[(1, 7), (2, 6), (0, 8)]
```

```python
#2

def zero_count(arr):
    count = 0
    result = []

    for num in arr:
        if num == 0:
            count = count+1
        else :
            result.append(num)

    result.extend([0]*count)
    return result, count

arr = [1,4,0,3,0,5,0,55]
result, count = zero_count(arr)
print("Array after moving zeros to the right :",result)
print("Number of zeros :",count)
```

```
Array after moving zeros to the right : [1, 4, 3, 5, 55, 0, 0, 0]
Number of zeros : 3
```

```python
#3

def intersection_arr(arr1, arr2):
    set1 = set(arr1)
    set2 = set(arr2)

    intersection = set1.intersection(set2)

    return list(intersection)
```

```
arr1 = [1,3,5,7,9,10]
arr2 = [2,4,6,8,10]
print(intersection_arr(arr1,arr2))
```

```
[10]
```